

Module 4 Assignment – SQL JOINS

Instructions

- Write SQL Server queries for the following business requirements.
- Choose the appropriate JOIN based on the requirement.
- Optimize your query for readability.

Business Scenario

A **Hospital Management System** maintains information about patients, doctors, appointments, departments, and medical records in separate tables.

Your task is to generate different reports for hospital management.

Assignment Questions

Q1. Display every patient along with the doctor assigned to them.

(Choose the appropriate JOIN.)

Q2. Display all patients, including those who have not yet been assigned a doctor.

Q3. Display all doctors, including those who currently have no patients assigned.

Q4. Generate a report displaying all patients and all doctors, regardless of whether they are assigned to each other.

Q5. Display patients who belong to the **Cardiology** department.

Q6. Display doctors who currently have **more than three patients** assigned.

Q7. Generate a report displaying:

- Patient Name
- Doctor Name
- Department Name
- Appointment Date

(Use Multi-Table JOINS.)

Q8. Display departments that currently have **no doctors assigned**.

Q9. Generate a complete Hospital Dashboard report displaying:

- Patient Name
- Doctor Name
- Department
- Appointment Date
- Medical Record Number

(Use all relevant tables.)

Q10. A doctor has recently joined the hospital but has not been assigned any patients.

Which JOIN would you use to display this doctor in your report? Explain your answer.

Q11. A patient has registered but has not yet booked an appointment. Write a query to identify such patients.

Q12. The hospital wants to count the total number of patients assigned to each doctor.

Display:

- Doctor Name
- Total Patients

Sort the report in descending order.

Q13. Explain the difference between the following JOINS using the hospital database as an example:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN

Q14. The hospital wants to create a report showing every possible combination of:

- Doctor
- Consultation Room

Which JOIN would you use?

Write the SQL query.